

# Manual Test Plan — Speech

**Covers:** kuantorflow#268 (the speech helper and the rate setting) and #272 (*Listen and type*) ·

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## Why this exists

The pytest suite cannot hear anything, and a headless browser generally has no voices at all. Everything either side of the audio is covered automatically — the setting sanitises and clamps, the round grades server-side, the page carries the right markup — but **whether a word is actually spoken, in the right voice, at the right speed, is only ever established by a person with speakers on**.

That is not a theoretical gap. Two bugs in this area reached a merged branch with the whole suite green, and both were found by playing the app:

- the availability probe read `window.kfSpeech` before the deferred module had loaded, so every browser was told it had no English voice;
- the Settings slider snapped back to the middle on reopening, because `form.reset()` restores page-load defaults.

Neither was visible to a test that checks markup. Assume this class of bug is what you are hunting.

## Before you start

Sign in — the rate slider is deliberately dead for anonymous visitors (#102), so a signed-out session cannot exercise most of section A.

Two things worth knowing in the browser console:

- `window.kfSpeech.voices()` prints the voice list the module chose from.
- `document.body.dataset.speechRate` is the value `speech.js` reads on every press.

## A. The rate setting (#268)

| #  | Do this                                                  | Expect                                                                                                                            |
|----|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| A1 | Set 50%, save, press a speaker                           | Noticeably slower.  <i>confirmed</i>         |
| A2 | Change the rate and press again <b>without reloading</b> | New speed applies at once.  <i>confirmed</i> |
| A3 | Set 150%, save, press                                    | Faster, and still intelligible                                                                                                    |
| A4 | Save 50%, close the popup, reopen it                     | Slider <b>and</b> the number both read 50%                                                                                        |
| A5 | Drag to 80%, close <b>without</b> saving, reopen         | Shows the saved value, not 80%                                                                                                    |
| A6 | Save 6 in <i>Cards per round</i> , reopen the popup      | Shows 6, not 10 — same bug class as A4                                                                                            |
| A7 | Sign out, open Settings                                  | Slider greyed and unmovable                                                                                                       |

|    |                                       |                                         |
|----|---------------------------------------|-----------------------------------------|
| A8 | Set a rate, then hard-reload the page | Rate survives; it is stored per account |
|----|---------------------------------------|-----------------------------------------|

## B. Listen and type (#272)

| #  | Do this                                                                     | Expect                                                  |
|----|-----------------------------------------------------------------------------|---------------------------------------------------------|
| B1 | Open a round                                                                | Play buttons and boxes only — <b>no words on screen</b> |
| B2 | Press play                                                                  | The word is spoken                                      |
| B3 | Press play four times quickly                                               | You hear it once more, not four times over              |
| B4 | Load the round and wait, touching nothing                                   | Silence. Nothing may speak without a press              |
| B5 | Type answers with capitals, a trailing full stop, a hyphen typed as a space | All marked correct                                      |
| B6 | Type a homophone ( <i>their</i> for <i>there</i> )                          | Marked <b>wrong</b> — this is the exercise              |
| B7 | Submit                                                                      | Results show each word with its meaning or translation  |
| B8 | Press a speaker on the results page                                         | Speaks, at the rate set in Settings                     |
| B9 | Change the rate mid-round, return, press play                               | New speed, without restarting the round                 |

## C. The speaker button elsewhere (#283)

The rate lives inside `speak()`, so **every** caller should obey it. Each of these was only tested as a mechanism, not per surface.

| #  | Do this                                               | Expect                        |
|----|-------------------------------------------------------|-------------------------------|
| C1 | Card deck — press the speaker on a card               | Speaks at the set rate        |
| C2 | Card deck — press the speaker                         | The card must <b>not</b> flip |
| C3 | Flashcard list — press a speaker                      | Speaks at the set rate        |
| C4 | Look up a word, press the speaker in the review popup | Speaks at the set rate        |

## D. Browsers and devices

Voice availability is a property of the browser, not of the app, and it is the one thing the server genuinely cannot know.

| #  | Where                | Expect                                                      |
|----|----------------------|-------------------------------------------------------------|
| D1 | Chrome               | Speaks; <code>voices()</code> lists an en-GB or en-US voice |
| D2 | Edge                 | Speaks                                                      |
| D3 | Firefox              | Speaks, or shows the no-voice panel if none are installed   |
| D4 | iPhone / iPad Safari | First <b>press</b> speaks. Nothing autoplays                |

|    |                                                                           |                                                                                                                              |
|----|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| D5 | Any browser with no English voice                                         | <i>Listen and type</i> shows its panel and offers other activities; speaker buttons disappear rather than sitting there dead |
| D6 | An English word while the browser's default voice is Ukrainian or Russian | Read by an English voice, not through Cyrillic phonology                                                                     |

D5 is hard to stage deliberately. A Linux machine with no speech packages is the usual way; failing that, it is the one case worth leaving to production.

## E. Judgement calls — only you can decide these

- **Is 150% still usable, or gibberish?** If gibberish, the ceiling is wrong and should come down.
- **Does 50% distort the word?** Some engines stretch phonemes badly below about 70%. If so, the floor should come up.
- **Is dictation at a slow rate a fairer test, or a different one?** A word slowed to half speed is arguably not the word a learner will meet in conversation.
- **Are multi-word expressions good listening questions?** *Take for granted* was kept deliberately, on the theory that catching the unstressed middle is the harder and better skill. Worth confirming it does not just sound mangled.

## F. Known limits — not bugs

- **Two tabs.** Save a rate in one; the other keeps its old value until reloaded. The value is rendered per page load, and a cross-tab sync was judged not worth the machinery.
- **The word is in the page source.** `speechSynthesis` needs the text in the browser, so a learner who opens view-source can read the answers. Inherent rather than an oversight; nothing in the round is scored or recorded.
- **Voice quality varies wildly.** This is a listening exercise, not a pronunciation reference — a dictionary is better for the latter, and the guide says so.

## Recording what you find

Report a failure by its number, plus the browser and what `window.kfSpeech.voices()` printed. That list is almost always the fastest way to narrow a speech bug, because it separates "*the app chose badly*" from "*the browser had nothing to choose from*".